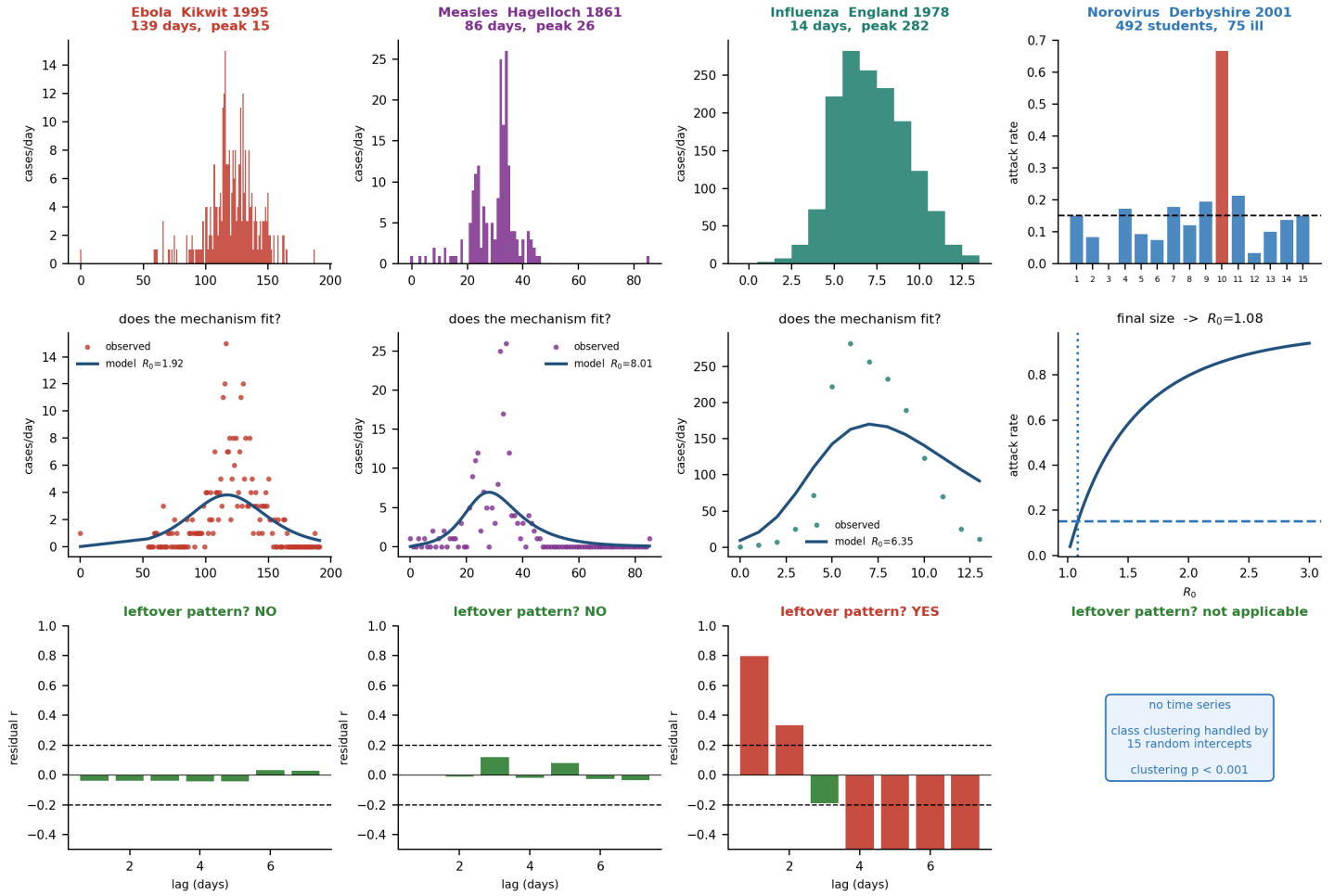


Pre-fit Diagnostics

Four outbreaks, checked before any model was fitted



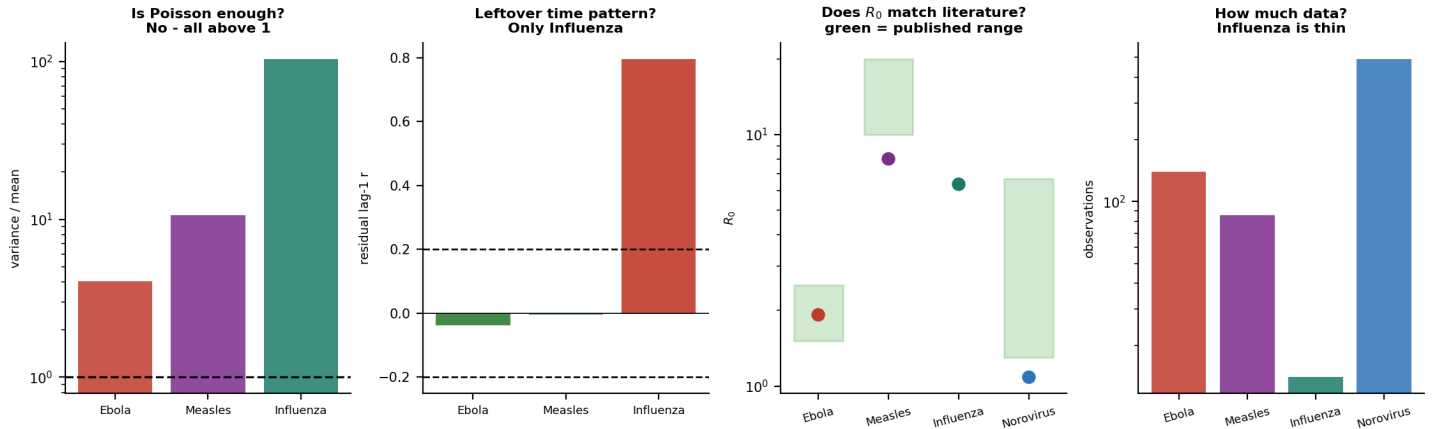
Verdict per outbreak

Outbreak	Variance/mean	Residual lag-1	Needs extra time term?	Status
Ebola	4.1	-0.037	no	clean
Measles	10.6	-0.005	no	clean
Influenza	103.6	+0.796	YES	pattern left over
Norovirus	n/a (binary)	n/a	n/a	clean

Top row is the raw data. Middle row overlays the mechanistic model. Bottom row asks whether any pattern is left in the errors — green bars mean the model captured the timing and nothing systematic remains.

Cross-outbreak Comparison

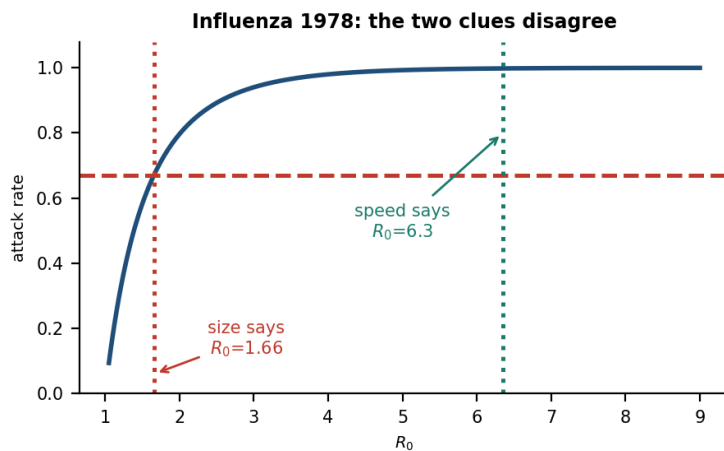
The four checks that decide how each arm is modelled



Modelling decisions that follow

Outbreak	What was counted	Observation model	Time-varying term	Verdict	Why
Ebola	daily new cases	Negative Binomial	no	PASS	clean fit, R_0 in published range
Measles	daily new rash	Negative Binomial	no	PASS	clean fit, complete village cohort
Norovirus	ill / not ill	Bernoulli + class effects	n/a	PASS	tight estimate, clustering handled
Influenza	students in bed	Negative Binomial	saturated - off	CAVEAT	only 14 days; two clues disagree

The one problem worth knowing about



Clue	Implied R_0
Final size (67% infected)	1.66
Epidemic speed	6.35
Published (Avilov 2024)	8.14

Influenza 1978 is a famous dataset that classic models cannot fully fit: the final size and the epidemic speed point to different R_0 values, and no published model reconciles them. It is kept as a stress test, not as a headline estimate. Ebola, Measles and Norovirus carry the comparison.

How to proceed

Next step	Reason
Report Ebola, Measles, Norovirus as the comparison	all three pass every pre-fit check
Report Influenza only with its caveat stated	documented conflict; cite Avilov et al. (2024)
Run once with and once without Influenza	confirms the caveated arm is not pulling the others
Compare each posterior interval to the green bands above	the interval, not the point estimate, is the claim